← Back **Graded quiz - Data and Lean Six Sigma**
Graded Assignment • 30 min

Due Mar 16, 11:59 PM CET

1. What is **not** a reason for organisations to improve their processes? 1 point

- Technological innovation.
- To gather more data.
- To withstand competition.
- To adapt to higher customer expectations.

2. What is DMAIC? 1 point

- DMAIC prescribes five phases to collect data.
- DMAIC is a data analytics project.
- DMAIC is a project to facilitate development of new ideas.
- DMAIC is a stepwise approach for process improvement.

3. Which DMAIC phase is about "Make a diagnosis of the current situation"? 1 point

- Define.
- Measure.
- Analyze.
- Improve.
- Control.

4. In a webshop, customers complain that they have to wait too long before their order arrives. A Lean Six Sigma project is started to reduce these complaints. 1 point

What is an appropriate CTQ for this project?

- Number of orders.
- Delivery date.
- Lead time.
- Customer satisfaction.

5. A densist treats patients according to a pre-set schedule. A project is started to decrease the patients waiting times. 1 point

What is an appropriate (observational) unit?

- Patient.
- Minutes.
- Dentist.
- Waiting time.

6. A teacher collects the ages for students from two different classes (class 1 and class 2). The table displays the data. 1 point

Age	Class
23	1
17	1
21	1
21	2
30	2
18	2

What is the **measurement unit** for the variable Age?

- Years.
- Student.
- 1 or 2.
- Number.

7. You are sitting in a lecture. 1 point

Which selection of students does **not** give a random sample?

- All students who come in late are selected.
- All students toss a coin and all students with heads are selected.
- Using Minitab ten names are sampled from a column with all the names of the students.
- All students roll a dice and all students who throw a 1 are selected.

8. What makes a sample a good sample? 1 point

- Is has to be representative.
- It should depend on the research question.
- It has to be very large.
- It has to consist of the whole population.

9. After collecting your data, but before analyzing your data, you should first organize your data. How should this be done in order to analyze your data with a statistical software package (like Minitab)? 1 point

- Units in the rows and variables in the columns.
- CTQs in the columns and variables in the rows.
- CTQs in the rows and variables in the columns.
- Units in the columns and variables in the rows.

10. 1 point

Call time	Employee	Type of call	Day
7.0	A	Question	Tuesday
5.1	B	Question	Tuesday
5.7	B	Remark	Wednesday
8.0	A	Complaint	Thursday
9.1	B	Other	Thursday

What is true about the displayed data?

- It shows 20 units distributed over 4 variables.
- It shows of 4 units and 5 variables.
- It shows 20 units distributed over 5 variables.
- It shows of 5 units and 4 variables.